

Predicting airline acceptance of fuel-saving reroutes from flight-plan revisions

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Abstract

Flight-plan revisions provide direct comparisons for learning route preferences: a route previously filed for a flight and its replacement. We use 1.48 million pairs from eighteen months of European traffic to train a ranking model with route indicators, flight context and waypoint identifiers. Shared flight attributes prevent message timing from revealing the label. On a later test quarter, the model identifies the replacement in 68.1 % of cases, compared with 55.0 % for the best single-indicator rule and 57.5 % for the strongest logistic baseline. Accuracy reaches 94.9 % on the fifth of pairs with the highest confidence. Matched examples of retained routes improve predictions on a smaller stay-pair test set. A retrospective policy then selects lower-fuel routes above a calibrated probability threshold. At an operating point targeting 80 % precision, 79.2 % of selections match the subsequent filing and account for 20.1 kilotonnes of planned-fuel differences over the quarter. These results establish useful prediction and confidence-based selection from revision records. The fuel total describes historical choices; acceptance of actual proposals and additional savings require prospective evaluation.

Keywords: air traffic flow management, airline preferences, learning to rank, fuel efficiency

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Declaration of competing interest

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CRedit authorship contribution statement

Ramon Dalmau: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft. **David Perez:** Conceptualization, Resources, Validation, Writing – review & editing. **Seddik Belkoura:** Methodology, Validation, Writing – review & editing. **Franck Ballerini:** Validation, Writing – review & editing. **Vincent Taverniers:** Resources, Data curation, Writing – review & editing. **Gratiel Marius Marin:** Resources, Validation, Writing – review & editing. **Robin Deransy:** Conceptualization, Project administration, Writing – review & editing. **Benjamin Cramet:** Supervision, Writing – review & editing. **Grigori Gustin:** Supervision, Project administration, Writing – review & editing.

Disclaimer

The views expressed are those of the authors and do not necessarily reflect the official position of EUROCONTROL or of its Member States. This paper reports research and does not commit to any operational service.

Data availability

The data used in this study is operational air traffic data held by EUROCONTROL and processed under existing data-handling arrangements; it cannot be made publicly available. Aggregated tables underlying every figure are available from the corresponding author upon reasonable request, in the de-identified form this paper itself uses: operators, city pairs and waypoints appear in them by rank rather than by name, which leaves every number and every ordering unchanged. Per-flight tables are excluded from the release.